



Modelling Financial Markets

Seminar 10: Market Microstructure — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ EKOP: Easley–Kiefer–O'Hara–Paperman (model of informed trading)
- ▣ ETF: Exchange-Traded Fund
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ MA: Moving Average
- ▣ OLS: Ordinary Least Squares
- ▣ PIN: Probability of INformed trading
- ▣ SE: Standard Error
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VECM: Vector Error-Correction Model
- ▣ VIX: CBOE Volatility IndeX

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the moments and the standard error of the Roll estimator, and its bias under autocorrelated order flow
- ▣ Derive the Glosten–Milgrom spread for any prior and the Kyle equilibrium; read λ as a regression slope
- ▣ Solve Almgren–Chriss by Euler–Lagrange; write a numerically stable PIN likelihood
- ▣ Do inference on real data: bootstrap, HAC, Monte Carlo under the null, break tests, IV, VECM price discovery, predictive-regression bias

- ▣ Background: Lecture 10; textbooks Hasbrouck (2007), Bouchaud et al. (2018), Franke, Härdle & Hafner (2019)

Structure

- ▣ **Part A:** ten derivations (A1–A10)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** an open question, a seed for the team project (C1), and the critique of an AI answer (C2)

How to Work in This Seminar

- ▣ **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- ▣ Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- ▣ Units: relative spreads and price moves in basis points ($1 \text{ bp} = 0.01\%$); traded value in USD
- ▣ Intraday data have strong dependence within a day: resample whole days, not single bars

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (Roll moments and bias), A3 (Glosten–Milgrom for any prior), A5 (deriving Kyle), A7 (Almgren–Chriss by Euler–Lagrange), A9 (the PIN likelihood)
- Part B: B1 (the intraday U-shape), B2 (spread estimators at two frequencies, Monte Carlo of Roll), B4 (illiquidity and the VIX, break test)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A6 (model: A5), A8 (model: A7), A10 (model: A9)
- Part B: B3 (spreads across 25 assets; model: B2), B5 (Amihud on three markets; model: B2, B4), B6 (volume and price moves, IV; model: B4), B7 (the Bitcoin clock; model: B1), B8 (price discovery; model: B4)
- Part C: C1, open question and project seed; C2, critique of an AI answer

Setup: Python Environment and Data

-  **Open in Colab:** [chapter10_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- SPY 5-minute bars, regular session, October 2020 – September 2026; Bitcoin 5-minute bars, September 2024 – September 2026
- Daily open, high, low, close and volume for 7 US stocks, 11 BVB stocks and 7 crypto-assets; VIX; BET-TR (BET with dividends reinvested)
- BVB traded values converted from lei to USD at the BNR reference rate of each day

Formulas Used Today

- Roll bias and precision [Harris \(1990\)](#): $\mathbb{E}[\hat{\gamma}_1] \approx \gamma_1 - (\gamma_0 + 2\gamma_1)/T$, $T\text{Var}(\hat{\gamma}_1) \rightarrow \gamma_0^2 + 3\gamma_1^2$ (Gaussian MA(1))
- Roll [Roll \(1984\)](#): $\text{Var}(\Delta p) = \sigma^2 + 2c^2$, $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$, $s = 2c$
- Glosten–Milgrom [Glosten & Milgrom \(1985\)](#): $a = V_L + (V_H - V_L)P(V_H \mid \text{buy})$, $b = V_L + (V_H - V_L)P(V_H \mid \text{sell})$, Bayes' rule
- Kyle [Kyle \(1985\)](#): $\beta = \sigma_u/\sigma_v$, $\lambda = \sigma_v/(2\sigma_u)$, expected profit $\sigma_v\sigma_u/2$; PIN [Easley et al. \(1996\)](#): $\alpha\mu/(\alpha\mu + \varepsilon_b + \varepsilon_s)$
- Almgren–Chriss [Almgren & Chriss \(2001\)](#): $x_k = X \sinh(\kappa(T - t_k))/\sinh(\kappa T)$, $\cosh(\kappa T) = 1 + \tilde{\kappa}^2 \tau^2/2$, $\tilde{\kappa}^2 = \lambda\sigma^2/\tilde{\eta}$, $\tilde{\eta} = \eta - \gamma\tau/2$
- Amihud [Amihud \(2002\)](#): $\text{ILLIQ} = \frac{1}{D} \sum_d |r_d|/\text{DVOL}_d$

A1: The Roll Model: Moments, Estimator, Bias [Solved]

Task

□ Context and assumptions

- ▶ Efficient price: a random walk $m_t = m_{t-1} + \varepsilon_t$, with ε_t i.i.d., mean 0, variance σ^2 Roll (1984).
- ▶ Trade price: $p_t = m_t + c q_t$, where c is the half-spread and $q_t = +1$ (buy) or -1 (sell), independent of ε .
- ▶ Numbers for (b): a BVB stock trading around 20 lei, with daily price changes $\text{Var}(\Delta p) = 0.0520$ and $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -0.0081$ (lei²).

□ Sub-tasks

- ▶ (a) Assume the q_t are i.i.d. with $P(q_t = 1) = 1/2$. Derive $\text{Var}(\Delta p_t)$ and $\text{Cov}(\Delta p_t, \Delta p_{t-1})$ from the definition of Δp_t .
- ▶ (b) With the numbers above, compute c , the spread $s = 2c$ in lei and in % of the price, the efficient variance σ^2 , the share of $\text{Var}(\Delta p)$ due to the bounce, and the first-order autocorrelation ρ_1 .
- ▶ (c) Now let the trade signs be a Markov chain with $\text{Corr}(q_t, q_{t-k}) = \rho^k$ and no information content. Derive $\text{Cov}(\Delta p_t, \Delta p_{t-1})$ and the value the Roll estimator converges to for $c = 0.090$, $\rho = 0.3$.

□ Report and interpret

- ▶ The two moments, the five numbers of (b) and the limit of (c).
- ▶ Interpretation: does a positive ρ make the Roll estimator too high or too low, and why?

A1: solution [Solved]

Solution (1/2)

- Step 1. Differencing gives $\Delta p_t = \varepsilon_t + c(q_t - q_{t-1})$: the efficient innovation plus the change in the side of the trade.
- Step 2. Since ε and q are independent, $\text{Var}(\Delta p_t) = \sigma^2 + c^2 \text{Var}(q_t - q_{t-1}) = \sigma^2 + c^2(1 + 1) = \sigma^2 + 2c^2$.
- Step 3. $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = c^2 \text{Cov}(q_t - q_{t-1}, q_{t-1} - q_{t-2})$; the only common term is q_{t-1} , with sign $-$, so the covariance is $-c^2 \text{Var}(q_{t-1}) = -c^2$.
- Step 4 (b). $c = \sqrt{0.0081} = 0.090$ lei, $s = 2c = 0.18$ lei, i.e. $0.18/20 = 0.9\%$ of the price.
- Step 5 (b). $\sigma^2 = 0.0520 - 2c^2 = 0.0520 - 0.0162 = 0.0358$; bounce share $2c^2/\text{Var}(\Delta p) = 31.2\%$;
 $\rho_1 = -0.0081/0.0520 = -0.156$.

A1: solution [Solved]

Solution (2/2)

- ▣ Step 6 (c). With $\text{Cov}(q_t, q_{t-k}) = \rho^k$: $\text{Cov}(q_t - q_{t-1}, q_{t-1} - q_{t-2}) = \rho - \rho^2 - 1 + \rho = -(1 - \rho)^2$.
- ▣ Step 7 (c). Hence $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2(1 - \rho)^2 = -0.0081 \cdot 0.49 = -0.0040$.
- ▣ Step 8 (c). The Roll estimator converges to $\sqrt{c^2(1 - \rho)^2} = c(1 - \rho) = 0.063$ lei instead of 0.090: 30% too low.
- ▣ Interpretation: almost a third of the daily variance is bid-ask noise, so volatility from trade prices overstates efficient volatility.
- ▣ Interpretation: persistent order flow (orders split into several trades) makes reversals rarer, hides part of the bounce and biases the Roll spread downwards.

A2: The Standard Error of the Roll Estimator [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Same population moments as A1(b): $\gamma_0 = \text{Var}(\Delta p) = 0.0520$, $\gamma_1 = \text{Cov}(\Delta p_t, \Delta p_{t-1}) = -0.0081$, so $c = 0.090$ and $s = 0.18$ lei [Harris \(1990\)](#).
- ▶ The estimator is $\hat{s} = 2\sqrt{-\hat{\gamma}_1}$, with $\hat{\gamma}_1$ the sample autocovariance from T price changes.
- ▶ Use the Gaussian approximation (Bartlett's formula for an MA(1)): $T \text{Var}(\hat{\gamma}_1) \rightarrow \gamma_0^2 + 3\gamma_1^2$.

□ Sub-tasks

- ▶ (a) Apply the delta method to $g(\gamma_1) = 2\sqrt{-\gamma_1}$ and write $\text{se}(\hat{s})$ as a function of γ_0 , γ_1 and T .
- ▶ (b) Evaluate $\text{se}(\hat{\gamma}_1)$, $\text{se}(\hat{s})$ and $\text{se}(\hat{s})/s$ for $T = 78$ (one day of 5-minute bars) and $T = 250$ (one year of daily data).
- ▶ (c) Approximate $P(\hat{\gamma}_1 > 0)$ for both T with the Normal distribution, and explain why this approximation is only rough here.

□ Report and interpret

- ▶ A table with T , $\text{se}(\hat{\gamma}_1)$, $\text{se}(\hat{s})$, relative error and $P(\hat{\gamma}_1 > 0)$.
- ▶ Interpretation: how often would a true Roll market show a positive covariance on a single day?

□ Model: A1

A3: Glosten–Milgrom for Any Prior [Solved]

Task

□ Context and assumptions

- ▶ The asset is worth V_L or V_H , $\Delta V = V_H - V_L$, with prior $P(V = V_H) = \theta$ [Glosten & Milgrom \(1985\)](#).
- ▶ One trader arrives per period: informed with probability μ (buys if $V = V_H$, sells otherwise), uninformed otherwise (buys or sells with probability $1/2$).
- ▶ A competitive risk-neutral market maker sets $a = \mathbb{E}[V \mid \text{buy}]$ and $b = \mathbb{E}[V \mid \text{sell}]$. Numbers for (c): $V_L = 90$, $V_H = 110$, $\mu = 0.3$.

□ Sub-tasks

- ▶ (a) Use Bayes' rule to derive $P(V_H \mid \text{buy})$ and $P(V_H \mid \text{sell})$, then the spread $a - b$ as a function of θ , μ and ΔV ; show that it is largest at $\theta = 1/2$.
- ▶ (b) Show that the expected price of the next trade equals $\mathbb{E}[V \mid \mathcal{F}_t]$ and that this expectation is a martingale.
- ▶ (c) Compute the ask, the bid and the spread at $\theta = 0.5$, the new quotes after one buy, and the spread at $\theta = 0.7$ from your formula.

□ Report and interpret

- ▶ The spread formula, the martingale argument and the quotes of (c).
- ▶ Interpretation: why is the spread largest when the market maker is most uncertain?

A3: solution [Solved]

Solution (1/2)

- Step 1. $P(\text{buy} \mid V_H) = \mu + (1 - \mu)/2 = (1 + \mu)/2$ and $P(\text{buy} \mid V_L) = (1 - \mu)/2$.
- Step 2. Bayes, with $D = 2\theta - 1$: $P(V_H \mid \text{buy}) = \theta(1 + \mu)/(1 + \mu D)$ and $P(V_H \mid \text{sell}) = \theta(1 - \mu)/(1 - \mu D)$.
- Step 3. $a - b = \Delta V [P(V_H \mid \text{buy}) - P(V_H \mid \text{sell})] = 4\mu\theta(1 - \theta) \Delta V / (1 - \mu^2(2\theta - 1)^2)$.
- Step 4. With $x = 2\theta - 1$, $4\theta(1 - \theta) = 1 - x^2$ and the spread is $\mu\Delta V(1 - x^2)/(1 - \mu^2x^2) \leq \mu\Delta V$, with equality only at $x = 0$, i.e. $\theta = 1/2$.
- Step 5 (b). $P(\text{buy})a + P(\text{sell})b = \mathbb{E}[V \mid \mathcal{F}_t]$ by the law of total expectation; $\mathbb{E}[\mathbb{E}[V \mid \mathcal{F}_{t+1}] \mid \mathcal{F}_t] = \mathbb{E}[V \mid \mathcal{F}_t]$ by iterated expectations, so beliefs and expected trade prices are martingales.

A3: solution [Solved]

Solution (2/2)

- Step 6 (c). At $\theta = 0.5$: $P(V_H \mid \text{buy}) = 0.65 \cdot 0.5 / (0.65 \cdot 0.5 + 0.35 \cdot 0.5) = 0.6500$, so $a = 90 + 20 \cdot 0.6500 = 103.00$; symmetrically $b = 97.00$; spread $6.00 = \mu \Delta V$.
- Step 7 (c). After one buy, $\theta = 0.6500$ and $P(V_H \mid \text{buy}) = 0.7752$: $a = 105.50$, $b = 100.00$, spread 5.50 .
- Step 8 (c). At $\theta = 0.7$: $4 \cdot 0.3 \cdot 0.21 \cdot 20 / (1 - 0.09 \cdot 0.16) = 5.11$, the same as the quotes computed directly.
- Interpretation: at $\theta = 1/2$ a single trade moves beliefs most, so adverse selection costs most; once beliefs lean one way, less remains to be learned and the spread narrows.

A4: How Fast Does the Market Maker Learn? [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Glosten–Milgrom as in A3 with $V_L = 90$, $V_H = 110$ and $\theta_0 = 1/2$.
- ▶ The true value is V_H and, to isolate the speed of learning, assume every trade is a buy.

□ Sub-tasks

- ▶ (a) Show that each buy multiplies the odds $\theta/(1 - \theta)$ by $(1 + \mu)/(1 - \mu)$, and that the bid after k buys equals $V_L + \Delta V \theta_{k-1}$.
- ▶ (b) Derive the smallest number of buys k after which the bid exceeds 108, and evaluate it for $\mu = 0.1$, 0.3 and 0.5.
- ▶ (c) Compare with the simulated learning paths of the lecture (Section 6), where uninformed traders also sell.

□ Report and interpret

- ▶ The formula for k and the three values.
- ▶ Interpretation: how does the speed of price discovery depend on the share of informed traders?

□ Model: A3

A5: Deriving the Kyle Equilibrium [Solved]

Task

▣ Context and assumptions

- ▶ One auction. The value is $v \sim N(p_0, \sigma_v^2)$; one insider observes v and submits a market order x [Kyle \(1985\)](#).
- ▶ Noise traders submit $u \sim N(0, \sigma_u^2)$, independent of v ; the market maker sees only $y = x + u$ and sets $p = \mathbb{E}[v \mid y]$.
- ▶ Numbers: $\sigma_v = 2$ USD, $\sigma_u = 10\,000$ shares; a second session with $\sigma_u = 20\,000$ shares.

▣ Sub-tasks

- ▶ (a) Conjecture $p = p_0 + \lambda y$. Derive the insider's optimal x from the first-order condition and write β in $x = \beta(v - p_0)$.
- ▶ (b) Given $x = \beta(v - p_0)$, derive λ by the projection theorem, solve the two equations for β and λ , and compute the expected insider profit.
- ▶ (c) Compute β , λ , the profit and the price change for $y = 15\,000$; then compare λ and the profit in the second session.

▣ Report and interpret

- ▶ The equilibrium formulas and the numbers of (c).
- ▶ Interpretation: why does more noise trading help the insider?

A5: solution [Solved]

Solution (1/2)

- Step 1. Given v , the expected profit is $\mathbb{E}[(v - p_0 - \lambda x - \lambda u)x \mid v] = (v - p_0)x - \lambda x^2$ (since $\mathbb{E}[u] = 0$).
- Step 2. First-order condition: $(v - p_0) - 2\lambda x = 0$, so $x = (v - p_0)/(2\lambda)$ and $\beta = 1/(2\lambda)$.
- Step 3. (v, y) is jointly Normal, so $\mathbb{E}[v \mid y] = p_0 + \frac{\text{Cov}(v, y)}{\text{Var}(y)}y$ with $\lambda = \frac{\beta \sigma_v^2}{\beta^2 \sigma_v^2 + \sigma_u^2}$.
- Step 4. Substituting $2\lambda\beta = 1$: $\beta^2 \sigma_v^2 + \sigma_u^2 = 2\beta^2 \sigma_v^2$, so $\beta = \sigma_u/\sigma_v$ and $\lambda = \sigma_v/(2\sigma_u)$.
- Step 5. Expected profit $\mathbb{E}[(v - p_0)x - \lambda x^2] = \beta \sigma_v^2 - \lambda \beta^2 \sigma_v^2 = \sigma_v \sigma_u/2$.

A5: solution [Solved]

Solution (2/2)

- ▣ Step 6 (c). $\beta = 10\,000/2 = 5\,000$ shares per USD of mispricing; $\lambda = 2/20\,000 = 0.0001$ USD per share.
- ▣ Step 7 (c). Expected profit $2 \cdot 10\,000/2 = 10\,000$ USD; for $y = 15\,000$, $\Delta p = 0.0001 \cdot 15\,000 = 1.50$ USD.
- ▣ Step 8 (c). With $\sigma_u = 20\,000$: $\lambda = 0.00005$ (half), $\beta = 10\,000$ and profit $20\,000$ USD (double).
- ▣ Interpretation: more noise makes the order flow less informative, so the market maker moves prices less per share; the insider trades more aggressively and earns more, while half of the information still enters the price.

A6: Kyle's Lambda as a Regression Slope [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ The Kyle economy of A5 ($\sigma_v = 2$ USD, $\sigma_u = 10\,000$ shares) with $p_0 = 50$ USD, repeated over many independent auctions.
- ▶ The notebook simulates 200 000 auctions to check the population results numerically.

□ Sub-tasks

- ▶ (a) Show that the population OLS slope of $p - p_0$ on y , and of $v - p_0$ on y , both equal λ ; give $\text{Var}(y)$.
- ▶ (b) Derive what the Amihud-type ratio $|r|/(p_0|y|)$, with $r = (p - p_0)/p_0$, estimates, and express it in bp per USD 1 million.
- ▶ (c) Real data give the total volume $|x| + |u|$ instead of the net flow $|y|$: derive the direction of the bias and quantify it by simulation.

□ Report and interpret

- ▶ The three results and the simulated ratios.
- ▶ Interpretation: what does the Amihud ratio measure in a Kyle world, and why is it biased?

□ Model: A5

A7: Almgren–Chriss by Euler–Lagrange [Solved]

Task

- Context and assumptions
 - ▶ Sell $X = 10^6$ shares within $T = 5$ days; parameters of the numerical example of [Almgren & Chriss \(2001\)](#): $\sigma = 0.95$ USD per day^{1/2}, temporary impact $\eta = 2.5 \cdot 10^{-6}$, permanent impact $\gamma = 2.5 \cdot 10^{-7}$, risk aversion $\lambda = 2 \cdot 10^{-6}$.
 - ▶ The trader minimises $\mathbb{E}[C] + \lambda \text{Var}[C]$, where C is the implementation cost.
- Sub-tasks
 - ▶ (a) In continuous time, minimise $\int_0^T (\eta \dot{x}^2 + \lambda \sigma^2 x^2) dt$ subject to $x(0) = X$, $x(T) = 0$: write the Euler–Lagrange equation, solve it and give κ_c .
 - ▶ (b) In discrete time with $N = 5$ daily steps ($\tau = 1$), compute $\tilde{\eta} = \eta - \gamma\tau/2$, $\tilde{\kappa}^2 = \lambda\sigma^2/\tilde{\eta}$, κ from $\cosh(\kappa\tau) = 1 + \tilde{\kappa}^2\tau^2/2$, and the holdings x_k .
 - ▶ (c) Compute $\mathbb{E}[C]$ and the standard deviation of C , and compare them with selling evenly ($\lambda = 0$).
- Report and interpret
 - ▶ $x(t)$, κ_c and κ , the schedule, and the cost–risk comparison.
 - ▶ Interpretation: what does the trader buy with the extra expected cost?

A7: solution [Solved]

Solution (1/2)

- Step 1. With $F(x, \dot{x}) = \eta \dot{x}^2 + \lambda \sigma^2 x^2$, Euler–Lagrange $\frac{d}{dt} \partial F / \partial \dot{x} = \partial F / \partial x$ gives $2\eta \ddot{x} = 2\lambda \sigma^2 x$, i.e. $\ddot{x} = \kappa_c^2 x$.
- Step 2. $\kappa_c^2 = \lambda \sigma^2 / \eta = 2 \cdot 10^{-6} \cdot 0.9025 / (2.5 \cdot 10^{-6}) = 0.722$, $\kappa_c = 0.8497$.
- Step 3. The general solution $Ae^{\kappa_c t} + Be^{-\kappa_c t}$ with $x(0) = X$, $x(T) = 0$ gives $x(t) = X \sinh(\kappa_c (T - t)) / \sinh(\kappa_c T)$; e.g. $x(1) = 427\,151$, $x(2) = 181\,712$.
- Step 4 (b). $\tilde{\eta} = 2.5 \cdot 10^{-6} - 1.25 \cdot 10^{-7} = 2.375 \cdot 10^{-6}$; $\tilde{\kappa}^2 = 2 \cdot 10^{-6} \cdot 0.9025 / (2.375 \cdot 10^{-6}) = 0.76$.
- Step 5 (b). $\cosh \kappa = 1 + 0.76/2 = 1.38$, so $\kappa = 0.8463$ and $\sinh(5\kappa) = 34.40$.

A7: solution [Solved]

Solution (2/2)

- ▣ Step 6 (b). $x_k = X \sinh(\kappa(5 - k)) / \sinh(5\kappa)$: $x = (1\,000\,000; 428\,599; 182\,933; 76\,296; 27\,643; 0)$.
- ▣ Step 7 (b). Daily sales $n_k = x_{k-1} - x_k$: $(571\,401; 245\,666; 106\,637; 48\,652; 27\,643)$, i.e. 57% on day 1.
- ▣ Step 8 (c). $\mathbb{E}[C] = \frac{1}{2}\gamma X^2 + \frac{\tilde{\eta}}{\tau} \sum n_k^2 = 1.078$ million USD and $\text{sd}[C] = \sigma \sqrt{\tau \sum x_k^2} = 0.449$ million USD.
- ▣ Step 9 (c). Selling evenly gives $\mathbb{E}[C] = 0.600$ and $\text{sd}[C] = 1.041$ million USD.
- ▣ Interpretation: front-loading pays about 0.48 million USD more in expected impact to cut the standard deviation of the cost by more than half; the continuous and discrete κ are close because $\kappa\tau$ is small.

A8: A More Risk-Averse Seller [Proposed]

Solution discussed in class

Task

- Context and assumptions
 - ▶ Same order and parameters as A7, but a ten times higher risk aversion, $\lambda = 2 \cdot 10^{-5}$.
- Sub-tasks
 - ▶ (a) Compute $\tilde{\kappa}^2$, the discrete κ , the continuous κ_c and the holdings x_k for $N = 5$.
 - ▶ (b) Explain why the discrete and continuous κ now differ much more than in A7.
 - ▶ (c) Compute $\mathbb{E}[C]$ and $\text{sd}[C]$ and the ratios to A7.
- Report and interpret
 - ▶ The schedule, the two κ and the cost–risk ratios.
 - ▶ Interpretation: when would a trader choose this schedule?
- **Model:** A7

A9: The PIN Likelihood, Step by Step [Solved]

Task

□ Context and assumptions

- ▶ EKOP model [Easley et al. \(1996\)](#): each day an information event occurs with probability α ; it is bad news with probability δ .
- ▶ Daily buys B and sells S are Poisson: uninformed rates $\varepsilon_b, \varepsilon_s$; informed traders add μ to the buys on good-news days and to the sells on bad-news days.
- ▶ Numbers: $\alpha = 0.3$, $\delta = 0.5$, $\mu = 400$, $\varepsilon_b = \varepsilon_s = 1\,000$; one day with $B = 1\,450$ and $S = 1\,000$.

□ Sub-tasks

- ▶ (a) Write the likelihood of one day as a three-component Poisson mixture, and the probability of informed trading PIN.
- ▶ (b) Compute PIN and explain why evaluating the likelihood directly in double precision fails for this day.
- ▶ (c) Factor out the common terms, derive a numerically stable log-likelihood, evaluate it with log-sum-exp, and compute the posterior probability of a good-news day.

□ Report and interpret

- ▶ PIN, $\ln L$ and the posterior probability.
- ▶ Interpretation: why are days classified almost surely when counts are large?

A9: solution [Solved]

Solution (1/2)

- Step 1. With $\pi_n(r) = e^{-r} r^n / n!$: $L = (1 - \alpha)\pi_B(\varepsilon_b)\pi_S(\varepsilon_s) + \alpha\delta\pi_B(\varepsilon_b)\pi_S(\mu + \varepsilon_s) + \alpha(1 - \delta)\pi_B(\mu + \varepsilon_b)\pi_S(\varepsilon_s)$.
- Step 2 (b). $\text{PIN} = \alpha\mu / (\alpha\mu + \varepsilon_b + \varepsilon_s) = 120 / 2\,120 = 0.0566$.
- Step 3 (b). In double precision e^{-1000} underflows to 0 and 1000^{1450} overflows to ∞ , so $0 \cdot \infty$ returns NaN and the optimiser fails or stops at a boundary [Lin & Ke \(2011\)](#).

A9: solution [Solved]

Solution (2/2)

- Step 4 (c). Factor out $K = e^{-\varepsilon_b - \varepsilon_s} (\mu + \varepsilon_b)^B (\mu + \varepsilon_s)^S / (B! S!)$ and set $x_b = \varepsilon_b / (\mu + \varepsilon_b)$, $x_s = \varepsilon_s / (\mu + \varepsilon_s)$.
- Step 5 (c). Then $L = K [(1 - \alpha)x_b^B x_s^S + \alpha\delta e^{-\mu} x_b^B + \alpha(1 - \delta)e^{-\mu} x_s^S]$; $\ln K$ uses $\ln B!$ from the log-gamma function.
- Step 6 (c). Take the log of each bracket term and combine them with log-sum-exp: $\ln L = -11.71$.
- Step 7 (c). The posterior probability of a state is its term's share of the bracket: good news 1.0000.
- Interpretation: with 450 extra buys against a Poisson standard deviation of about 32, the day is classified almost surely; this is why PIN is identified from the share of such days, not from single days.

A10: Reading Days with the PIN Model [Proposed]

Solution discussed in class

Task

- Context and assumptions
 - ▶ Same EKOP parameters as A9: $\alpha = 0.3$, $\delta = 0.5$, $\mu = 400$, $\varepsilon_b = \varepsilon_s = 1\,000$.
- Sub-tasks
 - ▶ (a) For a day with $B = 1\,000$ and $S = 1\,420$, compute the stable log-likelihood and the posterior probability of bad news.
 - ▶ (b) For a quiet day with $B = S = 1\,000$, compute the posterior probability of no news.
 - ▶ (c) Compute PIN when μ doubles to 800, and explain why PIN remains hard to estimate for very liquid stocks even with the stable likelihood.
- Report and interpret
 - ▶ The two posterior probabilities, $\ln L$ and the new PIN.
 - ▶ Interpretation: can an order imbalance always be read as private information?
- **Model:** A9

B1: The Intraday U-Shape of SPY [Solved]

Task

▣ Context and assumptions

- ▶ SPY 5-minute bars (data/market/intraday/SPY.US_5m.csv), regular session 09:30–16:00 New York time, only full days with 78 bars: 1 477 days, October 2020 – September 2026.
- ▶ Returns are log changes of the close within a day; the first bar's return runs from the day's open to its close. $|r|$ is in basis points.

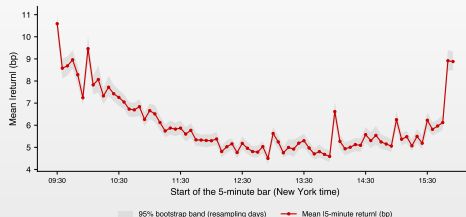
▣ Sub-tasks

- ▶ (a) Compute the mean $|r|$ of the first bar (09:30) and of the midday bars (11:30–14:00) over all days, and their ratio.
- ▶ (b) Compute the average share of daily volume traded in the last six bars (30 minutes).
- ▶ (c) Build 95% percentile bootstrap intervals for both statistics by resampling whole days with replacement (1 000 resamples, seed 42) [Efron \(1979\)](#).

▣ Report and interpret

- ▶ The two point estimates with intervals and the chart of mean $|r|$ by bar with its bootstrap band.
- ▶ Interpretation: why resample days rather than bars; is the U-shape systematic; what does it imply for a trader who must buy 1 million shares today?

B1: solution [Solved]



Solution

- Step 1. Mean $|r|$: 10.6 bp in the first bar and 5.1 bp at midday; ratio 2.06, 95% CI [1.98, 2.15].
- Step 2. Last 30 minutes: 18.6% of daily volume, 95% CI [18.3, 19.0]%, against $6/78 = 7.7\%$ if volume were even.
- Interpretation: bars of one day share its volatility and news, so resampling days keeps that dependence; the narrow intervals show a systematic pattern, and a large buyer should avoid the open and use the close or the closing auction.

B2: Spread Estimators at Two Frequencies [Solved]

Task (1/2)

□ Context and assumptions

- ▶ SPY 5-minute bars (`data/market/intraday/SPY.US_5m.csv`), regular session 09:30–16:00 New York time, only full days with 78 bars: 1 477 days, October 2020 – September 2026.
- ▶ Daily SPY bars from `data/market/SPY.US.csv` (prices adjusted for dividends), September 2024 – September 2026. One tick is USD 0.01.

B2: Spread Estimators at Two Frequencies [Solved]

Task (2/2)

□ Sub-tasks

- ▶ (a) Compute the Roll, Corwin–Schultz and Abdi–Ranaldo estimators within each day from the 5-minute bars, average them over days, and give 95% day-bootstrap intervals (1 000 resamples) [Roll \(1984\)](#), [Corwin & Schultz \(2012\)](#), [Abdi & Ranaldo \(2017\)](#).
- ▶ (b) Compute the same estimators from the daily bars, with a moving-block bootstrap (blocks of 20 days, 1 000 resamples).
- ▶ (c) Compare every estimate with one tick relative to the price.
- ▶ (d) Simulate the Roll model with SPY's bar volatility and 78 prices a day (1 477 days per sample, 200 samples), for a one-tick spread and for $c = 0$; compute the share of days with $\hat{\gamma}_1 > 0$ and the pooled estimate [Harris \(1990\)](#).

□ Report and interpret

- ▶ A table of estimates with intervals, and the Monte Carlo results next to the observed ones.
- ▶ Interpretation: which frequency gives a credible spread; what do the daily estimators measure; is the share of positive covariances evidence against the Roll model?

B2: solution [Solved]

Solution (1/2)

- Step 1 (a). 5-minute bars: Roll 2.82 bp [2.09, 3.43], Corwin–Schultz 3.11 bp [3.03, 3.20], Abdi–Ranaldo 1.06 bp [0.00, 1.67].
- Step 2 (b). Daily bars: Roll 64.3 bp, Corwin–Schultz 22.4 bp [17.4, 29.4], Abdi–Ranaldo 55.1 bp [0.0, 92.6].
- Step 3 (c). One tick is 0.21 bp: even the 5-minute Corwin–Schultz estimate is 15 times larger; the daily estimators are two orders of magnitude too high, because the daily standard deviation is 103.4 bp.
- Step 4. Abdi–Ranaldo is the square root of a mean that is often negative (set to 0), so its interval touches 0; the 5-minute Roll covariance is negative on only 59% of days.

B2: solution [Solved]

Solution (2/2)

- Step 5 (d). One-tick spread ($c = 0.10$ bp, bar s.d. 9.2 bp): $\hat{\gamma}_1 > 0$ on 45.3% of days [43.0, 48.1]; pooled Roll 2.07 bp [1.51, 2.53].
- Step 6 (d). No spread ($c = 0$): 45.3% [43.1, 48.3] and 2.06 bp [1.50, 2.53]: indistinguishable from a one-tick spread.
- Step 7 (d). Observed: 40.8% positive and 2.82 bp; bias-corrected Roll 1.92 bp.
- Interpretation: only intraday bars give credible magnitudes; daily estimators measure volatility. Positive days are not evidence against Roll: at $c/\sigma \approx 0.01$ the test has no power, and the 5-minute Roll estimate is mostly demeaning bias.

B3: Spread Estimates Across 25 Assets [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Daily open, high, low, close and volume, September 2024 – September 2026, from data/market: 7 US stocks and ETFs, 11 BVB stocks and the BET ETF group, 7 crypto-assets (25 assets).
- ▶ Volatility annualised with 252 days for stocks and 365 for crypto; the Amihud ratio as in the lecture (bp per USD 1 million).

□ Sub-tasks

- ▶ (a) Compute Corwin–Schultz and Abdi–Rinaldo for each asset from daily bars, and the median of each group.
- ▶ (b) Compute the Spearman rank correlations of Corwin–Schultz with annualised volatility, of the Amihud ratio with the median traded value, and of Corwin–Schultz with Amihud.
- ▶ (c) Count the assets for which Abdi–Rinaldo is set to 0.

□ Report and interpret

- ▶ Group medians, the three correlations and the scatter plots.
- ▶ Interpretation: do BVB stocks look more or less liquid than US stocks; which measure would you trust to compare markets; what would you need to validate it?

□ Model: B2

B4: Illiquidity and the VIX [Solved]

Task

□ Context and assumptions

- ▶ SPY 5-minute bars (data/market/intraday/SPY.US_5m.csv), regular session 09:30–16:00 New York time, only full days with 78 bars: 1 477 days, October 2020 – September 2026.
- ▶ Daily intraday illiquidity $ILLIQ_d$: the mean over the 78 bars of $|r|$ divided by the traded value, in bp per USD 100 million; VIX daily close from data/market/VIX.INDX.csv; 1 477 common days.

□ Sub-tasks

- ▶ (a) Estimate by OLS $\ln ILLIQ_d = a + b \ln VIX_d + e_d$.
- ▶ (b) Report OLS and HAC (Newey–West, 20 lags) standard errors, the 95% HAC interval, and the first-order autocorrelation of the residuals [Newey & West \(1987\)](#).
- ▶ (c) Add a linear time trend (in years) and re-estimate b with HAC errors.
- ▶ (d) Test the stability of b : interact $\ln VIX$ with a dummy for the second half of the sample (date fixed before estimation) and run a HAC Wald test at the 5% level.

□ Report and interpret

- ▶ $\hat{b}$ with both standard errors, the trend regression and the break test.
- ▶ Interpretation: read b as an elasticity; why do the standard errors differ; is the elasticity stable; is the relation causal?

B4: solution [Solved]

Solution

- ▣ Step 1 (a). $\hat{b} = 0.95$, $R^2 = 0.30$: a 10% higher VIX goes with about $0.95 \times 10\%$ higher illiquidity, nearly proportional.
- ▣ Step 2 (b). OLS SE 0.038, HAC SE 0.121 (3.2 times larger); 95% HAC CI [0.72, 1.19]; residual autocorrelation 0.76.
- ▣ Step 3 (c). With a trend: $\hat{b} = 0.80$ (SE 0.120), trend -0.062 per year (SE 0.021): illiquidity falls over time, but the VIX link remains.
- ▣ Step 4 (d). Break on 2023-09-27: $b = 1.03$ before and 0.60 after; difference -0.43 (HAC SE 0.25), Wald $p = 0.089 > 0.05$: stability is not rejected at 5%.
- ▣ Interpretation: persistent residuals make OLS errors far too small; the elasticity is close to 1 and weaker, but not significantly, in the second half; the relation is not causal, since both respond to news and uncertainty.

B5: Amihud Illiquidity on Three Markets [Proposed]

Solution discussed in class

Task

- ▣ Context and assumptions
 - ▶ The 25 assets of B3; daily Amihud ratio $|r_d|/DVOL_d$ in bp per USD 1 million, averaged over September 2024 – September 2026 [Amihud \(2002\)](#).
 - ▶ BVB traded value converted to USD at the BNR reference rate of each day.
- ▣ Sub-tasks
 - ▶ (a) Compute the median Amihud ratio of each group and the ratio BVB median / US median.
 - ▶ (b) Build a 95% bootstrap interval for that ratio by resampling assets within each group (1 000 resamples).
 - ▶ (c) Compute the Spearman correlation between the asset rankings of the first and the second year.
- ▣ Report and interpret
 - ▶ The medians, the ratio with its interval and the rank correlation.
 - ▶ Interpretation: how much less liquid is the BVB, how precise is that number, and is illiquidity a stable characteristic?
- ▣ **Model:** B2, B4

B6: Volume and Price Moves in SPY Bars [Proposed]

Solution discussed in class

Task (1/2)

☐ Context and assumptions

- ▶ SPY 5-minute bars (`data/market/intraday/SPY.US_5m.csv`), regular session 09:30–16:00 New York time, only full days with 78 bars: 1 477 days, October 2020 – September 2026.
- ▶ Relative volume = bar volume / median volume at the same time of day; $z = |r|$ divided by the day's standard deviation of 5-minute returns; 115 052 bars.

B6: Volume and Price Moves in SPY Bars [Proposed]

Solution discussed in class

Task (2/2)

□ Sub-tasks

- ▶ (a) Sort bars into 20 equal-size groups by relative volume, regress $\ln \bar{z}$ on $\ln$ mean relative volume, and give a 95% day-bootstrap interval for the slope (300 resamples).
- ▶ (b) At bar level, regress $\ln |r| - \ln \sigma_{\text{day}}$ on $\ln$ relative volume with standard errors clustered by day, and test slope = 0.5 at 5%.
- ▶ (c) IV (Instrumental Variable): instrument $\ln$ relative volume by the log relative volume of the same bar on the previous day; report the first stage, the two-stage least squares slope and its clustered SE.

□ Report and interpret

- ▶ The grouped and bar-level slopes, the t statistic against 0.5, and the IV results.
- ▶ Interpretation: is the relation concave, is it the square-root law [Tóth et al. \(2011\)](#), why cluster by day, and is the exclusion restriction of (c) plausible?

□ Model: B4

B7: The Bitcoin Clock [Proposed]

Solution discussed in class

Task

- ▣ Context and assumptions
 - ▶ Bitcoin 5-minute bars (`data/market/intraday/BTC-USD.CC_5m.csv`), UTC, September 2024 – September 2026: 749 days, trading 24 hours a day, 7 days a week.
- ▣ Sub-tasks
 - ▶ (a) On weekdays, compute the mean $|r|$ in 13:00–17:00 UTC (US morning) and in 03:00–11:00 UTC (Asian and European morning), and their ratio with a 95% day-bootstrap interval (1 000 resamples).
 - ▶ (b) Compute the ratio of weekday to weekend mean $|r|$ with a 95% day-bootstrap interval.
- ▣ Report and interpret
 - ▶ The two ratios with intervals.
 - ▶ Interpretation: why does a 24/7 market have a clock, and when would you execute a large Bitcoin order?
- ▣ **Model:** B1

B8: Price Discovery, the BET ETF and the BET-TR Index [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Daily log closes of the Patria-TV BET ETF (data/market/TVBETETF.R0.csv, days with trades) and of BET-TR (data/market/BETTR.INDX.csv), common days, September 2024 – September 2026 (lecture, Section 10).
- ▶ The ETF reinvests dividends, so it should be cointegrated with BET-TR with vector $\beta = (1, -1)'$.

□ Sub-tasks

- ▶ (a) Run the Johansen trace test [Johansen \(1991\)](#), then estimate the VECM with $\beta = (1, -1)'$ imposed and lags chosen by BIC; test $\alpha_{\text{ETF}} = 0$ and $\alpha_{\text{idx}} = 0$ at 5%.
- ▶ (b) Compute the Hasbrouck information-share bounds [Hasbrouck \(1995\)](#) and the Gonzalo–Granger component share [Gonzalo & Granger \(1995\)](#), with 95% moving-block bootstrap intervals (blocks of 20 days, 1 000 resamples).
- ▶ (c) Re-estimate both α on each half of the sample.

□ Report and interpret

- ▶ The adjustment coefficients with t statistics and intervals, the share bounds, and the two half-samples.
- ▶ Interpretation: which venue leads and how sure are you; why are the bounds so wide at daily frequency; why can the component share leave $[0, 1]$?

□ Model: B4

C1: Does Illiquidity Carry a Premium on the BVB?

Task (1/2)

☐ Context and assumptions

- ▶ Open question and seed for the team project: how illiquid is the BVB compared with US and crypto markets, and is illiquidity priced?
- ▶ Amihud (2002): expected illiquidity should **raise** future returns; an unexpected rise should **lower** current returns [Amihud \(2002\)](#).
- ▶ Data: monthly BET-TR, SPY and Bitcoin returns from data/market; market illiquidity from the stocks (tokens) of each market, 2016–2026.

C1: Does Illiquidity Carry a Premium on the BVB?

Task (2/2)

□ Sub-tasks

- ▶ (a) Build a monthly market illiquidity index (cross-sectional mean of log Amihud, months with at least 10 trading days) and its shock, the residual of an AR(1).
- ▶ (b) Regress monthly returns on lagged illiquidity and the shock, with HAC standard errors (6 lags), and correct the lagged coefficient for the Stambaugh bias [Stambaugh \(1999\)](#) with the reduced-bias estimator [Amihud & Hurvich \(2004\)](#).
- ▶ (c) On the BVB, form the illiquid-minus-liquid half of the ten blue chips, rebalanced every year, and test its mean return.

□ Report and interpret

- ▶ A table per market, the scatter of returns on shocks, and a discussion of power: few stocks, ten years [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#).
- ▶ Interpretation: which of Amihud's two predictions survives on each market?

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

The student asks an AI assistant

- "I have daily closes P (lei) and volumes V (number of shares, positive every day) for a BVB stock. Estimate the Roll spread and the Amihud illiquidity. What does a round trip cost me?"

The AI answer: text

- Roll: $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$ and $s = 2c$ Roll (1984)
- If the covariance comes out positive, take its absolute value: the formula still gives the spread
- Amihud: the average of $|r_d|$ divided by the volume of day d , per 1 million Amihud (2002)
- Result: $s = 0.05$ lei on a 20-lei stock, i.e. 0.25% of the price; a round trip (buy, then sell) costs $2s$, i.e. 0.50%

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

The AI answer: the code that came with the text

```
import numpy as np
dp = P.diff().dropna().values
g = np.cov(dp[1:], dp[:-1])[0, 1]
s_roll = 2 * np.sqrt(abs(g))
s_pct = 100 * s_roll / P.mean()
r = np.log(P).diff().dropna()
illiq = 1e6 * (r.abs() / V.loc[r.index]).mean()
```

C2: Critique an AI Answer (3/3) [Proposed]

Solution discussed in class

Task

- ▣ (a) The answer contains three errors; everything else is correct. Find each error in the text and in the code.
- ▣ (b) For each error, state how you would detect it: a test, a published number or a formula from the lecture.
- ▣ (c) Give the corrected version of each error: the formula, the code line and the corrected number.
- ▣ (d) Log each error as in `AI_ERRORS.md`; interpretation: which error would change an investment decision?
 - ▶ What you asked, what the AI answered, what was wrong, how you found out, the correct version

Hints

- ▣ What sign of the autocovariance does the Roll model predict? What does the opposite sign tell you?
- ▣ Compare with the definition of $ILLIQ$ on the Formulas frame: what is $DVOL_d$?
- ▣ Draw the bid, the ask and the mid-price: at which prices do you buy and sell?

Common Pitfalls

- ▣ Reading a daily high-low spread estimate as the true spread of a liquid asset
- ▣ Comparing illiquidity across markets without converting the traded value to one currency
- ▣ Treating 5-minute bars of the same day as independent observations
- ▣ Using raw prices and volumes across a stock split: the traded value jumps
- ▣ Calling a volume–volatility relation in bars a price-impact law

Take-Home Assignment (Team Project)

- Choose three assets not covered in class (a BVB stock outside the ten blue chips, a crypto token, a US small cap)
- Compute Amihud and the two high-low spread estimators by month; plot them with the volatility
- Estimate the cost of selling a position of USD 1 million with the square-root law, using each asset's volume and volatility
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- Log every AI error you catch (at least three per project) in `AI_ERRORS.md` and declare your AI use in `AI_USE.md`

Further Study on Quantinar

Advanced courses on quantinar.com

-  Applied Computational Finance

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_10

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